

SEMESTER S6

VLSI VERIFICATION & TESTING

Course Code	PEEVT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCEVT402 Digital System Design	Course Type	5 Credit Elective

Course Objectives:

1. To understand the complexity of verification and build SoC verification environments.
2. To use UVM to develop scalable, reusable, and efficient verification environments for complex designs.
3. To gain proficiency in System Verilog for designing and verifying complex digital systems
4. To design and implement effective test strategies to detect and diagnose faults in VLSI circuits.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Scope of testing and verification in VLSI design process; Issues in testing and verification of complex chips; embedded cores and SOC's, Functional verification, Timing verification. System Level Verification: Test Plan-Block Level Verification-Interface Verification. Application Based Verification-Canonical SoC Design-Testbench for the Canonical Design. Fast Prototype vs. 100 Percent Testing - FPGA Prototyping, Emulation Based Testing, Silicon Prototyping. Gate Level Verification-Sign Off Simulation, Gate Level Simulation with Unit-Delay Timing , Gate Level Simulation with Full Timing. Formal Verification. Choosing Simulation Tools.	9

2	Specialized Hardware for System Verification - Accelerated Verification Overview, RTL Acceleration-Software Driven Verification, Traditional In Circuit Verification, Intellectual Property, Design Guidelines for Accelerated Verification. UVM: Introduction, Verification Planning and Coverage-Driven Verification, Multi-Language and Methodologies, UVM Overview, UVM Testbench and Environments, Interface UVCs, System and Module UVCs, Software UVCs, The SystemVerilog UVM Class Library, UVM Utilities. UVM Library basics, Interface UVCs, Automating UVC Creation, Simple Testbench Integration, Register and Memory Package, System UVCs and Testbench Integration.	9
3	SystemVerilog: Basic circuits design using SystemVerilog. Procedural Statements and Functions, Implementation of OOPs Concepts in System Verilog, SystemVerilog DPI (Direct Programming Interface), Example, Enum Cast, Code library, SystemVerilog TestBench-SystemVerilog TestBench and Its components, combinational circuit – TestBench Example, Memory Model – TestBench Example. Connecting the testbench and design. 4 port ATM Router- Case study.	9
4	Testing: VLSI testing process and test equipment, Automatic Test Equipment, SCOAP Controllability and Observability, High-Level Testability Measures, Combinational circuit test generation-Redundancy Identification (RID)-Combinational ATPG Algorithms-Test Generation Systems-Test Compaction, Sequential circuit test generation-ATPG for Single-Clock Synchronous Circuits-Simulation-Based Sequential Circuit ATPG, Memory Test, Delay Test, IDDQ test, Design for testability-DFT and Scan Design-Partial-Scan Design. Built-In-Self-Test (BIST)- Random Logic BIST, Memory BIST, Boundary Scan Standard, Analog Test Bus Standard.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Literature Review and Report (10 Marks)

Assessment Method:

- Students select recent publications on a specific topic related to the course (eg. BIST).
- Preparation of a report summarizing the findings, discussing the significance, and proposing future research directions.

Criteria:

- Relevance of Chosen Publications (2 Marks): Selection of up-to-date and significant research papers.
- Depth of Analysis (4 Marks): Thorough understanding and critical analysis of the literature.
- Clarity and Organization (2 Marks): Well-structured report with clear arguments.
- Originality (2 Marks): Innovative insights or perspectives.

2. UVM for verification and Report (5 Marks)

Assessment Method:

- Use UVM for verification of digital circuits(eg. ALU Verification with UVM, Asynchronous FIFO) using ModelSim Questa/ Synopsys VCS/ Cadence Incisive Enterprise Simulator/ Xilinx Simulator (XSIM) or any other FOSS tools.
- Preparation of a detailed report.

Criteria:

- Comprehensiveness (2 Marks): Detailed explanation of the design and the tool used.
- Application Understanding (1 Mark): Insight into how the designing relate to the theoretical concepts covered in class.
- Report Quality (1 Mark): Clear and concise presentation of information.
- Reflective Analysis (1 Mark): Personal reflections and insights gained from the realization.

3. Simulation Tool Familiarization for SystemVerilog and Report (5 Marks)**Assessment Method:**

- Students use/study simulation tools (e.g., using ModelSim Questa/ Synopsys VCS/ Cadence Incisive Enterprise Simulator/ Xilinx Simulator (XSIM) or any other FOSS tools) to model digital designs discussed in the course.
- They prepare a report on the simulations, including the methodology, results, and interpretations.

Criteria:

- Tool Knowledge (2 Marks): Understand the features of the simulation tool.
- Familiarisation of Simulations (1 Mark): Representation of design characteristics.
- Report Quality (2 Mark): Well-organized and clearly written report.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain proficiency in industry standards and methodologies for design and verification for VLSI	K2
CO2	Create, configure and customize reusable, scalable, and robust UVM Verification Components (UVCs)	K3
CO3	Analyse the use of procedural statements and routines in testbench design with system verilog.	K4
CO4	Apply OOP concepts in designing testbench with system verilog	K3
CO5	Effectively test VLSI systems using existing test methodologies, equipments and tools.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3		3							
CO2	3		3	3	3							
CO3	3	3	3		3							
CO4	3		3	3	3							
CO5	3		3	3	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	SystemVerilog for Verification	Chris Spears	Springer	2nd Edition, 2008
2	Digital Systems Testing and Testable Design	M. Abramovici, M. A. Breuer, A. D. Friedman	Piscataway, New Jersey: IEEE Press,	1994
3	Reuse Methodology Manual for System-on-a-Chip Designs	Micheal Keating & Pierre Bricaud	Springer	2002
4	A Practical Guide to Adopting Universal Verification Methodology (UVM)	Sharon Rosenberg & Kathleen A Meade	Lulu publishers (Lulu.com)	2nd Edition, 2012
5	The UVM Primer: A Step-by-Step Introduction to the Universal Verification Methodology	Ray Salemi	Boston Light Press	2013
6	Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits	M. Bushnell and V. D. Agarwal	Kluwer Academic Publishers	2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	IEEE Standard for SystemVerilog—Unified Hardware Design, Specification, and Verification Language, 10.1109/IEEESTD.2018.8299595	IEEE	IEEE	2018
2	Introduction to Formal Hardware Verification	T.Kropf	Springer	2000
3	System-on-a-Chip Verification-Methodology and Techniques	P. Rashinkar, Paterson and L. Singh	Kluwer Academic Publishers	2001
4	Principles of Testing Electronic Systems	Samiha Mourad and Yervant Zorian	Wiley	2000
5	SoC Verification Methodology and Techniques	Prakash Rashinkar & Peter Paterson	Springer	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1, 2, 4	Design Verification and Test of Digital VLSI Circuits, https://archive.nptel.ac.in/courses/106/103/106103116/
2, 3	Unleashing SystemVerilog and UVM Video Series, Synopsys https://m.youtube.com/playlist?list=PLEgCreVKPx5AP61Pu36QQE0Pkni2Vv-HD